

Accessibility in the Loop (AITL) Framework

Specification, Version 0.1 (Working Draft)

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1. PURPOSE

AITL is an evaluation tool for the accessibility of AI-generated user interfaces. It applies the AI-Generated Interface Accessibility Assurance Framework (AIAAF): the evaluation framework comprising the six dimension rubric (§4), the generative failure pattern taxonomy, the 0 to 5 scoring scale with its severity driven release gate (§5), and the element level evidence rules. AIAAF defines what “accessible” means for AI-generated design and how to score output against that definition; AITL is the process (§6) that applies AIAAF and converts the results into guidance that design teams and tool-makers can act on before release. AITL keeps expert and assistive-technology-user judgment central; it does not treat AI as an automatic fix.

2. SCOPE AND NON-GOALS

- In scope: user interfaces produced or substantially shaped by AI systems, generated screens, components, markup, and design specifications, evaluated against WCAG 2.2 (A/AA), Section 508, and documented assistive-technology behavior.
- Out of scope (v0.1): non-UI content generation (long-form text, media), native platform apps beyond web-rendered UI, and automated remediation. AITL evaluates and guides; it does not claim to auto-fix.
- AITL complements, it does not replace, automated checkers (axe, WAVE, Lighthouse) and formal conformance audits. Its distinct contribution is the layer those tools do not cover: contextual judgment and the failure patterns specific to generative systems.

3. INPUTS

- **The artifact** in whatever form is available: rendered interface, underlying markup or component structure, design specification, or the generative prompt that produced it. More forms yield a deeper evaluation; any one form is sufficient to begin.
- **The context of use:** the user task under evaluation; assistive technologies in scope (screen reader, keyboard only, magnification, switch access); and the conformance target (e.g., WCAG 2.2 AA).

4. THE AIAAF RUBRIC, SIX DIMENSIONS

Every evaluation scores the artifact on six dimensions: the four WCAG principles, extended with the two layers automated tools miss most. Each finding is recorded at the element level with an evidence-based rationale and a mapped success criterion.

D1, Perceivable

Text alternatives are meaningful, not merely present; contrast meets minimums (4.5:1 normal text, 3:1 large text and UI components); structure and relationships are programmatically conveyed; media has alternatives.

D2, Operable

Full keyboard operability without traps; visible focus with logical order; touch/click targets $\geq 24 \times 24$ CSS px; user control over motion, timing, and auto-updating content.

D3, Understandable

Persistent, programmatically associated labels; predictable behavior; errors identified in text, programmatically associated to their field, with recovery guidance; logical reading order.

D4, Robust

Valid, semantic markup; correct name, role, and value on all interactive elements; correct ARIA usage (no aria-hidden on focusable content); compatibility with assistive-technology parsing.

D5, Assistive-Technology Behavior

How the output actually performs under a screen reader and keyboard against the defined task: are elements announced sensibly, is the task completable, does the experience match the visual one? This is the contextual layer conventional checkers cannot assess and is verified by hands-on AT testing, ultimately including testing with assistive-technology users.

D6, Resistance to Generative Failure Patterns

The dimension specific to AI-generated design: a growing, versioned taxonomy of the defect classes generative systems characteristically produce. Initial taxonomy (v0.1):

- Placeholder only labeling, fields “look” labeled but expose no accessible name.
 - divAsButton, non-semantic containers with click handlers standing in for real controls.
 - Generic or hallucinated alternative text, alt values produced because the attribute exists, not from the image’s purpose.
 - Visual only structure, hierarchy implied by styling but absent from markup (styled text instead of headings).
 - Color only state, errors and states conveyed by color with no text equivalent.
 - Suppressed focus, outline resets for visual cleanliness with no replacement indicator.
 - Low-contrast default aesthetics, light-gray secondary text below contrast minimums.
 - Decorative ARIA misuse, aria-hidden or roles applied in ways that hide interactive content from AT.
- The taxonomy is intended to grow through application and to be validated and published; each entry records the pattern, why generative systems produce it, its detection cue, and its remediation.

5. SCORING

Each dimension is scored 0 to 5 with element level findings recorded beneath it. Findings carry a severity that drives the release gate.

Score	Label	Meaning
5	Conformant	No defects found on this dimension for the evaluated task; meets the conformance target.
4	Minor issues	Defects exist but do not impede task completion with assistive technology; low-severity fixes.
3	Degraded	Task completable with assistive technology but with significant friction or workarounds.
2	Barrier	At least one defect prevents some users from completing the task on this dimension.
1	Blocking	Multiple defects; the task is effectively unavailable to assistive-technology users.
0	Not evaluable	Artifact insufficient to assess this dimension; noted, not averaged.

Severity	Definition	Gate effect
Critical	Prevents task completion for screen reader	Fails the release gate.

	or keyboard only users.	
High	Major friction or exclusion for a user group; task completable with difficulty.	Fails gate unless documented exception.
Medium	Conformance failure with available workaround.	Must be scheduled for remediation.
Low	Best-practice deviation; no conformance failure.	Advisory.

6. PROCESS

- **Step 1, Frame:** define the task, AT in scope, and conformance target.
- **Step 2, Sweep:** run automated checks to clear the mechanically detectable layer.
- **Step 3, Evaluate:** score D1 to D6 element by element, with keyboard and screen reader passes for D5 and the taxonomy for D6; every finding gets a rationale grounded in specific evidence and a mapped criterion.
- **Step 4, Route:** separate what is decidable from what requires human or AT-user judgment, and route the latter explicitly.
- **Step 5, Return signals:** produce the score profile, prioritized findings, upstream prompt/specification guidance, and the gate determination.

7. OUTPUTS

- A six dimension score profile with a conformance determination against the stated target.
- A prioritized, element level findings list (criterion, severity, rationale, fix).
- Upstream guidance: prompt- and specification-level requirements that steer future generation away from the defects found.
- A release-gate result (Critical findings block).
- A documented trace suitable for conformance evidence and research use.

8. VALIDATION ROADMAP

- Phase 1 (current): apply the rubric to synthetic and publicly available AI-generated interfaces; refine dimension definitions and the D6 taxonomy. A completed worked example accompanies this specification.
- Phase 2: inter-rater exercises with accessibility practitioners to test that the rubric scores consistently across evaluators.
- Phase 3: validation sessions with assistive-technology users to test that rubric findings predict real task success and failure.
- Phase 4: dissemination through peer reviewed human-computer-interaction venues and professional channels; publication of the taxonomy for field adoption.

Phase 5 (vision): integration by AI tool makers. The long term aim, once the framework is validated, is adoption by the companies that build AI design and code generation tools, such as Anthropic, OpenAI, and others, so that AIAAF is integrated directly into how their tools produce interfaces. Through the MCP server and a published, machine readable taxonomy, a generation tool could consult the framework as it works and steer output away from the known failure patterns. This work builds on an active area of research and practice in accessibility and AI rather than starting from nothing. The practical goal is that AI generated UI arrives substantially more accessible by default, so designers and developers spend far less effort retrofitting accessibility afterward. Human and assistive technology user review remains part of the process for the judgments automation cannot make. This work builds on active research and practice across the field rather than starting from nothing; the contribution here is the framework's focus on AI generated user interfaces and its named taxonomy of the failure patterns those tools characteristically produce. No such collaboration exists yet; this describes the direction the endeavor is built to enable.

9. RELATIONSHIP TO EXISTING WORK

AITL builds on established standards (WCAG 2.2, Section 508, WAI-ARIA) and on the general “human-in-the-loop” evaluation tradition. Its specific contribution is narrower and deeper: an element level evaluation rubric purpose-built for AI-generated user interfaces, and a named, growing taxonomy of generative accessibility failure patterns, the layer that neither automated checkers nor generic conformance audits currently address.

10. VERSION HISTORY

- v0.1 (July 2026), initial specification: six dimension rubric, 0 to 5 scale, severity gate, eight-pattern D6 taxonomy, five-step process, validation roadmap. Companion artifact: AITL Worked Example (sign-up screen evaluation).